

35) 3×5 coefficient matrix w/ 3 pivot columns.
 $\uparrow \quad \uparrow$
 rows columns
 Is the system consistent?

3 rows $\rightarrow$ 3 equations
 5 columns $\rightarrow$ 5 unknowns (variables)

The system is consistent w/ infinitely many possible solutions. The fact that there are 3 pivot columns suggest that 2 of the 5 variables are free and the other 3 will have solutions that depend on one or more of the free variables.

37) Coefficient matrix w/ a pivot in every row.
 Size not specified.

Since each row has a pivot, each corresponding equation will not yield a contradictory statement, so there will always be at least one solution, or infinitely many, depending on the # of rows vs # of columns.

41) # equations $<$ # variables = underdetermined system
 $\uparrow \quad \uparrow$
 # rows # columns

* See explanation from #35. Not all variables will have a pivot position, so there will inevitably be some free variables.

43) # equations $>$ # variables = overdetermined system
 $\uparrow \quad \uparrow$
 # rows # columns

Yes, it can be consistent (or inconsistent).
 If the coefficient matrix has a pivot in every row, none of the equations will yield a contradiction, so the

system will have @ least one solution.
Any extra variables beyond the pivots would be free (rows of all 0's in rref) which will lead to infinitely many solutions.

one or more of the equations in an overdetermined system is redundant, as they do not add any new information to the system, these "extra" equations can be written as a combination of the other equations in the system, and they don't create a contradiction.

Ex:
$$\begin{cases} x+y=1 \\ 2x+2y=2 \\ x-y=0 \end{cases} \leftarrow \text{redundant equation}$$

Note it is a constant multiple of the 1st equation.

$$\begin{aligned} x &= 1-y \\ (1-y)-y &= 0 \\ 1-2y &= 0 \rightarrow y = \frac{1}{2} \end{aligned} \rightarrow x = \frac{1}{2} \quad \text{sol: } \left(\frac{1}{2}, \frac{1}{2}\right)$$

So the system is consistent w/ 1 solution.

Ex: Here's an example of an overdetermined inconsistent system (b/c it can happen too):

$$\begin{cases} x+y=1 \\ x-y=0 \\ 2x+y=5 \end{cases} \rightarrow x = \frac{1}{2}, y = \frac{1}{2}, \text{ but}$$

when we plug it into the 3rd equation:

$$2\left(\frac{1}{2}\right) + \frac{1}{2} = 1 + \frac{1}{2} = \frac{3}{2} \neq 5$$

Ex: Consistent ~ infinitely many solutions:

$$\begin{cases} x+y=1 \\ -x-y=-1 \\ 2x+2y=2 \end{cases} \leftarrow \text{Note that they are all constant multiples of each other!}$$

4b) t = velocity (independent variable)

Equation needs to satisfy:

$$(1) 0 = a_0 + a_1(0) + a_2(0)^2 + a_3(0)^3 + a_4(0)^4 + a_5(0)^5$$

$$(2) 2.90 = a_0 + a_1(2) + a_2(2)^2 + a_3(2)^3 + a_4(2)^4 + a_5(2)^5$$

$$(3) 14.8 = a_0 + a_1(4) + a_2(4)^2 + a_3(4)^3 + a_4(4)^4 + a_5(4)^5$$

$$(4) 39.6 = a_0 + a_1(6) + a_2(6)^2 + a_3(6)^3 + a_4(6)^4 + a_5(6)^5$$

$$(5) 74.3 = a_0 + a_1(8) + a_2(8)^2 + a_3(8)^3 + a_4(8)^4 + a_5(8)^5$$

$$(6) 119 = a_0 + a_1(10) + a_2(10)^2 + a_3(10)^3 + a_4(10)^4 + a_5(10)^5$$

So our augmented matrix will look like:

$$\left[\begin{array}{cccccc|c} 1 & 0 & 0^2 & 0^3 & 0^4 & 0^5 & 0 \\ 1 & 2 & 2^2 & 2^3 & 2^4 & 2^5 & 2.90 \\ 1 & 4 & 4^2 & 4^3 & 4^4 & 4^5 & 14.8 \\ 1 & 6 & 6^2 & 6^3 & 6^4 & 6^5 & 39.6 \\ 1 & 8 & 8^2 & 8^3 & 8^4 & 8^5 & 74.3 \\ 1 & 10 & 10^2 & 10^3 & 10^4 & 10^5 & 119 \end{array} \right] \leftarrow \text{plug this monster into a matrix calculator}$$

I got: $a_0 = 0$

$$a_3 = 0.66$$

$$a_1 = 1.71$$

$$a_4 = -0.07$$

$$a_2 = -1.19$$

$$a_5 = 0.002$$

* see attached printout from maple *

Also using maple: $p(7.5) = 64.83$

* A 3rd degree polynomial would not pass through all data points!
!!

↑
makes sense based on table :)

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> with(Student[LinearAlgebra]);
[&x, `', AddRow, AddRows, Adjoint, ApplyLinearTransformPlot, BackwardSubstitute, BandMatrix, (1)
  Basis, BilinearForm, CharacteristicMatrix, CharacteristicPolynomial, ColumnDimension,
  ColumnSpace, CompanionMatrix, ConstantMatrix, ConstantVector, CrossProduct,
  CrossProductPlot, Determinant, DeterminantSteps, Diagonal, DiagonalMatrix, Dimension,
  Dimensions, EigenPlot, EigenPlotTutor, Eigenvalues, EigenvaluesTutor, Eigenvectors,
  EigenvectorsTutor, Equal, GaussJordanEliminationTutor, GaussianElimination,
  GaussianEliminationTutor, GenerateEquations, GenerateMatrix, GramSchmidt,
  HermitianTranspose, HouseholderMatrix, Id, IdentityMatrix, IntersectionBasis, InverseTutor,
  IsDefinite, IsOrthogonal, IsSimilar, IsUnitary, JordanBlockMatrix, JordanForm,
  LUDecomposition, LeastSquares, LeastSquaresPlot, LinearSolve, LinearSolveTutor,
  LinearSystemPlot, LinearSystemPlotTutor, LinearTransformPlot, LinearTransformPlotTutor,
  MatrixBuilder, MatrixExponential, MatrixInverse, MinimalPolynomial, Minor, MultiplyRow,
  Norm, Normalize, NullSpace, Pivot, PlanePlot, ProjectionMatrix, ProjectionPlot,
  Pseudoinverse, QRDecomposition, RandomMatrix, RandomVector, Rank,
  ReducedRowEchelonForm, ReflectionMatrix, RotationMatrix, RowDimension, RowSpace,
  SetDefault, SetDefaults, SingularValues, SumBasis, SwapRow, SwapRows, Trace, Transpose,
  UnitVector, VectorAngle, VectorSumPlot, ZeroMatrix, ZeroVector]

```

```

> A :=

```

$$A := \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 2 & 2^2 & 2^3 & 2^4 & 2^5 & 2.90 \\ 1 & 4 & 4^2 & 4^3 & 4^4 & 4^5 & 14.8 \\ 1 & 6 & 6^2 & 6^3 & 6^4 & 6^5 & 39.6 \\ 1 & 8 & 8^2 & 8^3 & 8^4 & 8^5 & 74.3 \\ 1 & 10 & 10^2 & 10^3 & 10^4 & 10^5 & 119 \end{bmatrix};$$

```

A :=

```

$$A := \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 2 & 4 & 8 & 16 & 32 & 2.90 \\ 1 & 4 & 16 & 64 & 256 & 1024 & 14.8 \\ 1 & 6 & 36 & 216 & 1296 & 7776 & 39.6 \\ 1 & 8 & 64 & 512 & 4096 & 32768 & 74.3 \\ 1 & 10 & 100 & 1000 & 10000 & 100000 & 119 \end{bmatrix}$$

(2)

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> ReducedRowEchelonForm(A)

```

$$\begin{bmatrix} 1. & 0. & 0. & 0. & 0. & 0. & 0. \\ 0. & 1. & 0. & 0. & 0. & 0. & 1.712499980 \\ -0. & -0. & 1. & -0. & -0. & -0. & -1.194791663 \\ -0. & -0. & -0. & 1. & -0. & -0. & 0.6614583331 \\ -0. & -0. & -0. & -0. & 1. & -0. & -0.07005208333 \\ 0. & 0. & 0. & 0. & 0. & 1. & 0.002604166667 \end{bmatrix} \quad (3)$$

> LinearSolve(A)

$$\begin{bmatrix} 0. \\ 1.712499970 \\ -1.194791662 \\ 0.6614583331 \\ -0.07005208333 \\ 0.002604166667 \end{bmatrix} \quad (4)$$

> $p := t \mapsto 0 + 1.712499970 t - 1.194791662 t^2 + 0.6614583331 t^3 - 0.07005208333 t^4 + 0.002604166667 t^5;$

$p := t \mapsto 1.712499970 \cdot t - 1.194791662 \cdot t^2 + 0.6614583331 \cdot t^3 - 0.07005208333 \cdot t^4 + 0.002604166667 \cdot t^5$ (5)

> $p(7.5);$

64.83837891 (6)

>